

VERAVAL, India

WMO: 429090

Lat: 20.900N

Lon: 70.367E

Elev: 26

StdP: 14.68

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	59.0	61.1	28.7	22.7	68.7	34.2	28.9	69.6	17.7	76.4	16.2	76.7	5.9	50	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	7.2	94.8	74.4	93.0	77.9	91.7	79.3	84.7	90.0	84.0	89.3	83.4	88.5	7.4	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.4	174.6	88.4	82.6	170.3	87.7	81.9	166.3	87.1	49.2	90.2	48.3	89.4	47.6	88.8	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.7	21.1	18.3	DB	54.0	102.1	2.6	1.9	52.1	103.5	50.6	104.6	49.1	105.7	47.3	107.1	(4)
(5)			WB	44.7	85.7	3.1	1.2	42.4	86.6	40.7	87.3	38.9	88.0	36.7	88.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.0	72.8	74.6	79.0	82.5	85.8	86.9	84.1	82.4	83.0	84.3	81.0	75.8		
	DBStd	5.02	2.82	3.43	3.23	2.55	1.81	2.06	2.16	1.78	1.85	2.20	2.64	3.12		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	1	0	1	0	0	0	0	0	0	0	0	0	0		
	CDD50	11331	706	688	900	975	1109	1108	1057	1004	989	1065	930	800		
	CDD65	5857	241	269	435	525	644	658	592	539	539	600	480	335		
	CDH74	64087	1691	1923	3698	5729	8359	8934	7270	5923	6089	7042	4773	2656		
	CDH80	25329	359	487	1237	2112	3945	4626	2851	1598	1995	3169	2079	871		
(14) Wind	WSAvg	8.3	6.2	6.8	7.3	7.9	9.4	11.9	14.2	11.9	7.5	5.4	5.1	5.6		
(15) Precipitation	PrecAvg	32.2	0.0	0.0	0.0	0.0	0.1	6.7	13.2	6.9	3.6	0.7	0.3	0.0		
	PrecMax	74.3	0.4	0.8	0.5	0.4	1.5	36.6	60.3	25.2	12.5	6.8	6.4	0.4		
	PrecMin	6.6	0.0	0.0	0.0	0.0	0.0	0.0	0.7	0.4	0.0	0.0	0.0	0.0		
	PrecStd	15.8	0.1	0.1	0.1	0.1	0.2	6.5	9.5	6.3	3.5	1.3	1.0	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.1	90.6	95.9	97.9	93.4	94.1	90.7	88.8	91.3	98.2	94.8	90.7		
		MCWB	66.4	66.7	68.7	71.3	82.9	83.7	83.0	81.2	81.7	72.4	70.9	70.1		
	2%	DB	85.4	87.2	91.0	91.4	91.8	92.8	89.3	87.3	89.5	95.2	92.7	88.5		
		MCWB	66.3	68.6	71.1	76.7	82.8	83.6	82.4	80.9	81.1	74.8	72.3	70.0		
	5%	DB	83.7	84.9	88.0	89.1	91.0	91.7	88.2	86.1	88.3	93.2	91.1	86.8		
		MCWB	67.0	69.5	73.5	79.2	82.5	83.4	82.0	80.5	80.4	77.0	73.0	69.7		
	10%	DB	81.9	83.0	85.9	87.9	90.0	90.6	87.1	85.1	87.2	91.1	89.3	85.1		
		MCWB	67.1	69.8	74.4	79.2	82.0	83.1	81.6	80.2	80.0	78.3	73.3	69.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.3	77.6	80.7	83.3	85.3	85.7	84.6	82.8	84.3	84.6	82.5	78.2		
		MCDB	82.5	83.8	86.9	88.4	90.6	91.0	88.5	86.1	88.7	91.0	88.6	85.6		
	2%	WB	73.9	75.3	78.8	82.1	84.3	85.0	83.5	81.9	82.7	82.9	80.2	76.0		
		MCDB	81.3	82.6	85.2	87.3	90.1	90.3	87.5	85.3	87.3	89.4	87.2	84.0		
	5%	WB	72.0	73.7	77.4	81.0	83.5	84.3	82.7	81.3	81.7	81.9	78.6	74.4		
		MCDB	79.8	81.3	84.1	86.6	89.2	89.7	86.6	84.5	86.2	88.4	86.3	82.9		
	10%	WB	70.2	72.4	76.3	80.1	82.7	83.7	82.0	80.7	80.9	80.9	77.0	72.6		
		MCDB	78.9	80.1	83.0	85.7	88.4	88.9	85.8	83.9	85.2	87.3	85.3	81.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	20.3	18.4	16.1	12.8	9.5	7.2	4.9	4.9	8.2	15.2	18.5	19.7		
		MCDBR	21.3	20.5	19.3	14.7	9.8	8.7	6.4	6.2	10.3	18.8	19.8	20.8		
	5% WB	MCWBR	15.3	15.1	14.0	8.8	4.4	3.7	3.4	3.2	5.1	11.8	13.4	14.4		
		MCWBR	18.8	17.3	15.1	12.0	9.5	8.1	5.8	5.5	8.9	14.3	16.9	18.8		
(40) Clear-Sky Solar Irradiance	taub		0.406	0.410	0.463	0.498	0.549	0.699	0.740	0.667	0.570	0.540	0.516	0.442		
	taud		2.250	2.250	2.108	2.049	1.971	1.692	1.628	1.767	1.972	1.971	1.993	2.144		
	Ebn at Noon		272	279	268	260	245	209	200	216	237	237	235	255		
	Edh at Noon		40	43	51	55	59	77	82	72	58	56	52	44		
(44) All-Sky Solar Radiation	RadAvg		1555	1833	2137	2314	2334	1788	1262	1326	1648	1775	1552	1455		
	RadStd		44	49	41	40	76	158	145	134	140	77	72	68		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page